

Yousif Shasha

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Summary

Structural Engineering student seeking a 12-month Engineering Co-op with Aecon's Civil West division; preferred location: Toronto, ON. Academic experience in surveying, AutoCAD, Excel, and materials testing, plus fabrication with TMU's Concrete Canoe team. Supervisory experience coordinating tasks and training employees; interested in heavy civil construction.

Education

Toronto Metropolitan University

B.Eng. Structural Engineering

Toronto, ON

Expected 2029

- Relevant Coursework: Surveying, Project Management, Concrete Materials, Foundation and Soils Engineering, Civil Engineering Systems, Structural Design and Analysis, and Transportation Engineering.
- Awards: *Dean's List - Fall 2025, Winter 2026. Top 3 Project, Geology for Engineers*

Experience

Groupe Adonis

Grocery Supervisor

Mississauga, ON

July 2023 – Present

- Supervise 2-6 employees, coordinate daily tasks, and train new hires to maintain efficient workflows and teamwork.
- Manage inventory, stocking, palletizing, and organization of 10,000+ products to maintain availability.
- Problem-solve customer and operational issues using leadership, communication, and adaptability in a fast-paced setting.

Projects

Topographic Surveying — *Total Station, Excel*

October 2025

- Collected total-station elevation, angle, and distance measurements with a 0.0032 m benchmark misclosure.
- Calculated traverse components, checked closure, and applied coordinate adjustments.
- Checked field accuracy using direct/reverse observations and differential-levelling checks.

Structural Mechanics — *AutoCAD, Excel*

Fall 2025 - Winter 2026

- Produced AutoCAD drawings for aluminum, timber, and composite structural specimens.
- Performed hand and Excel calculations for stress, deflection, truss forces, and torsion.
- Compared theoretical results with strain-gauge and dial-gauge measurements, with a 1.14% neutral-axis difference.

Geotechnical Engineering Laboratory

Winter 2026

- Classified soils using Atterberg limits and sieve analysis to assess plasticity and gradation.
- Evaluated compaction, permeability, and consolidation; identified a maximum dry unit weight of 19.54 kN/m³.
- Assessed soil strength through direct shear and compression testing, measuring a 37.4° friction angle.

Hydraulics Engineering — *EPANET, HEC-RAS, Excel*

Winter 2026

- Modelled pipe-network flow and pressure in EPANET, varying roughness, diameter, demand, and redundancy.
- Investigated pressurized flow, pipe losses, weirs, and sluice-gate behaviour through seven laboratory experiments.
- Used HEC-RAS and Excel to compare flood profiles up to the 100-year event and assess velocity and erosion risk.

Leadership & Activities

Toronto Metropolitan University Concrete Canoe Team

Construction Team Member

Toronto, ON

September 2025 – Present

- Fabricated canoe components through precision cutting, sanding, taping, and assembly to meet competition requirements.
- Tested alternative materials and fabrication methods to improve structural performance and constructability.
- Documented prototype results and collaborated with team members to guide iterative design improvements.

Skills

Technical: AutoCAD, Microsoft Excel, Revit, ETABS (Basic), MATLAB, Python, HEC-RAS, EPANET, LaTeX

Engineering: Surveying, Engineering Materials Testing, Technical Reporting, Data Analysis, Uncertainty & Error Analysis, Structural Analysis, Strength of Materials, Hydraulics.

Soft: Problem Solving, Communication, Teamwork, Time Management, Adaptability, Leadership.